Prithvi Raj

Prithvi-R | Prithvi Raj | prithvi.raj23b@iiitg.ac.in | Portfolio

Education

Indian Institute of Information Technology, Guwahati Aug 2023 – Present
B.Tech in Electronics and Communication Engineering **CPI:** 7.73 **SPI:** 8.71

Research & Core Projects

Lock-Free Concurrent Queues (SPSC & MPMC) [Github] Aug 2026 - Oct 2026

- Designed and benchmarked a zero-allocation Single-Producer Single-Consumer (SPSC) queue and a Multi-Producer Multi-Consumer (MPMC) queues using explicit atomic operations.
- Achieved a 153x speedup over standard mutual exclusion (`std::mutex`) models by enforcing fine-grained (`memory_order_acquire`) and (`memory_order_release`) semantics.

Low-Latency Multimodal Desktop Assistant [Github] May 2026 - June 2026

- Engineered a lightweight, native C++ system utility utilizing Win32 API and GDI+ to capture screen buffers and audio streams for real-time multimodal inference.
- Designed an asynchronous multithreaded network worker using atomic queues and thread pools, achieving non-blocking API dispatch and zero UI frame drops.

Project Ascend: AI-Powered Life Tracker [Github] July 2025

- Programmed a full-stack wellness tracker using React, gamifying daily habits with AI-generated side-quests.
- Logged activities and generated weekly reports via a modular backend with JWT auth and PostgreSQL.

RoundUPX: Microservices Financial Savings Platform [Github] March 2025 - April 2025

- Architected scalable RESTful microservices in Go to process real-time UPI transaction round-ups, defining clean API contracts, schemas, and background job workers for asynchronous event handling.
- Integrated third-party payment APIs, implementing idempotency keys, exponential backoff retries, and transactional reconciliation logic to reliably manage webhook callbacks and edge-case status codes.
- Engineered automated CI/CD pipelines via GitHub Actions to run unit tests and deploy containerized Docker services to AWS cloud environments with minimal downtime.
- Instrumented structured logging and distributed tracing patterns across endpoints to streamline production debugging, partner response verification, and metric tracking.

Experience & Leadership

Semiconductor Workforce Development Program *TCFST & NTHU, Taiwan* March 2026

- Executed 2-micrometer precision multi-layer silicon wafer fabrication via hands-on cleanroom operations at National Tsing Hua University, achieving over 90% yield for critical layers.
- Designed and tested four new Photolithography processes, leading to a significant reduction in production waste by 30%; techniques are now adopted as standard practices across the fabrication teams in the collaboration.

Research Intern (SPARC Program) *IIT Hyderabad* Nov 2025

- Fabricated a functional Resistive Gas Sensor detecting different gas levels, specifically optimizing for Ammonia.
- Successfully executed multi-layer semiconductor fabrication with 10-micrometer detail.

Coordinator *Alankrit Fine-Arts Club – IIIT Guwahati* Dec 2024 – July 2025

- Led 5+ campus-wide arts events, managing logistics; recognized as the club's most active coordinator.

Technical Skills

Languages: C++, Go (Golang), Python, JavaScript, SQL
Systems & Backend: REST APIs, Idempotency & Retries, Background Jobs, Multi-threading, Win32 APIs
Cloud & Infrastructure: AWS, Docker, CI/CD Pipelines (GitHub Actions), PostgreSQL
AI & ML Frameworks: PyTorch, 3D Gaussian Splatting(3DGS), OpenCV, Scikit-learn, Pandas, NumPy

Achievements

Hackathon Finalist, India International Science Festival (IISF) 2024

- Recognized as a finalist team in the top 0.2% (from 45,000+ competing teams) in the national-level S&T Hackathon.

First Prize, GitHub University Blind Coding Challenge 2023

- Ranked #1 among all first-year students across every engineering discipline in a university-wide competition.